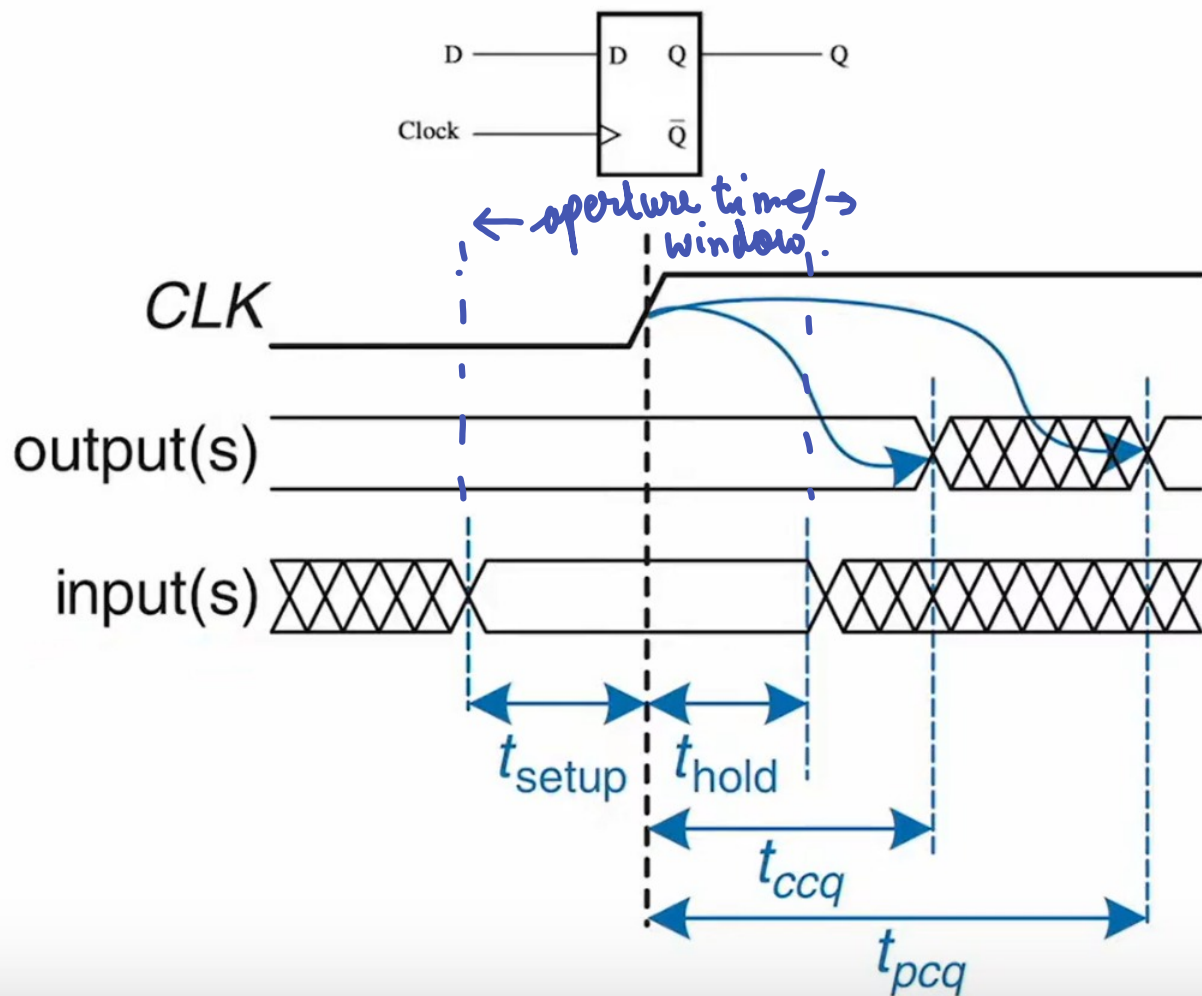


Timing Parameters

FF Timing Parameters



② Input to F.F. has to be stable for certain period of time around active edge of clock.

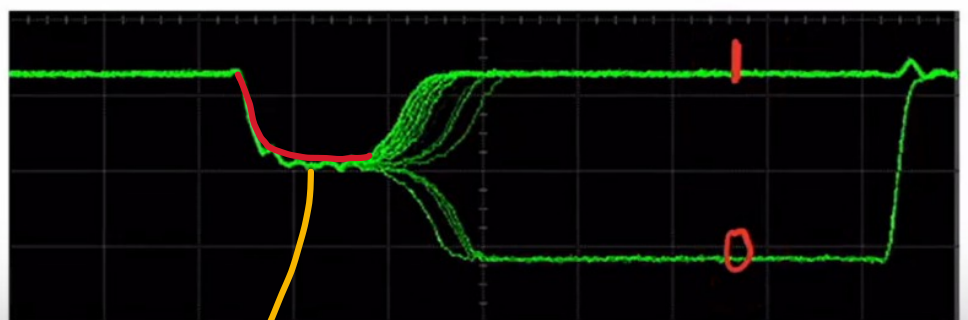
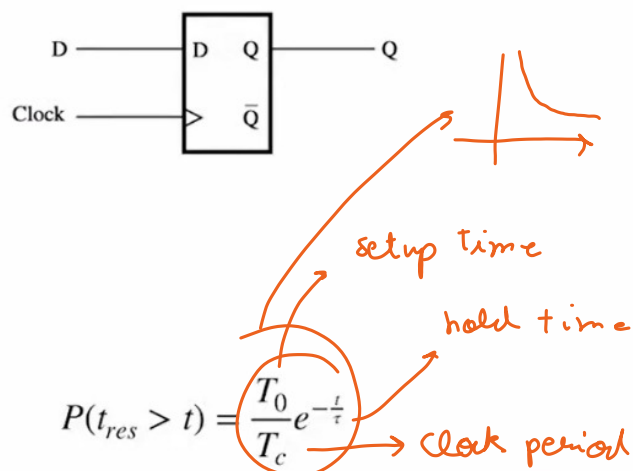
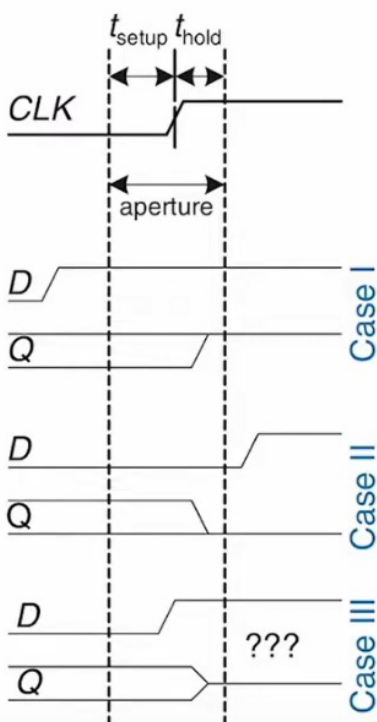
↳ before edge: t_{setup} .
after edge: t_{hold}

② t_{cq} (contamination delay to q) is min time & t_{pq} (propagation delay to q) is max time it may take to reflect the input

↳ it depends on how we construct the f.f. & type of transistors used while making.

Metastability

Metastability

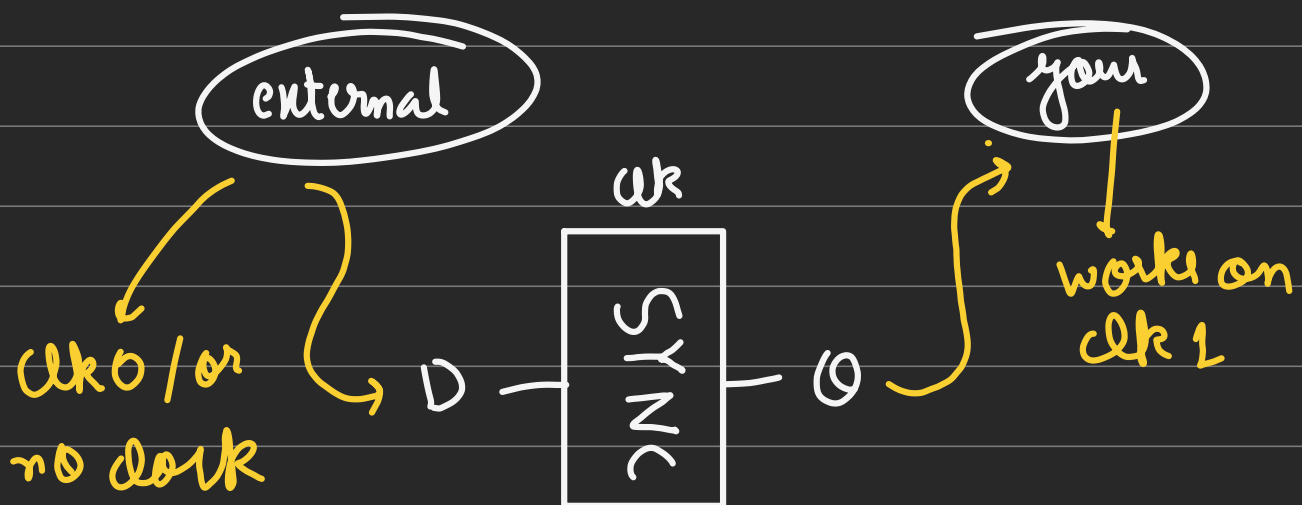


metastable stable.
bistable state

so the probability that it will stay in metastable state & not resolve to 1/0 goes to zero very quickly (exponential decay)

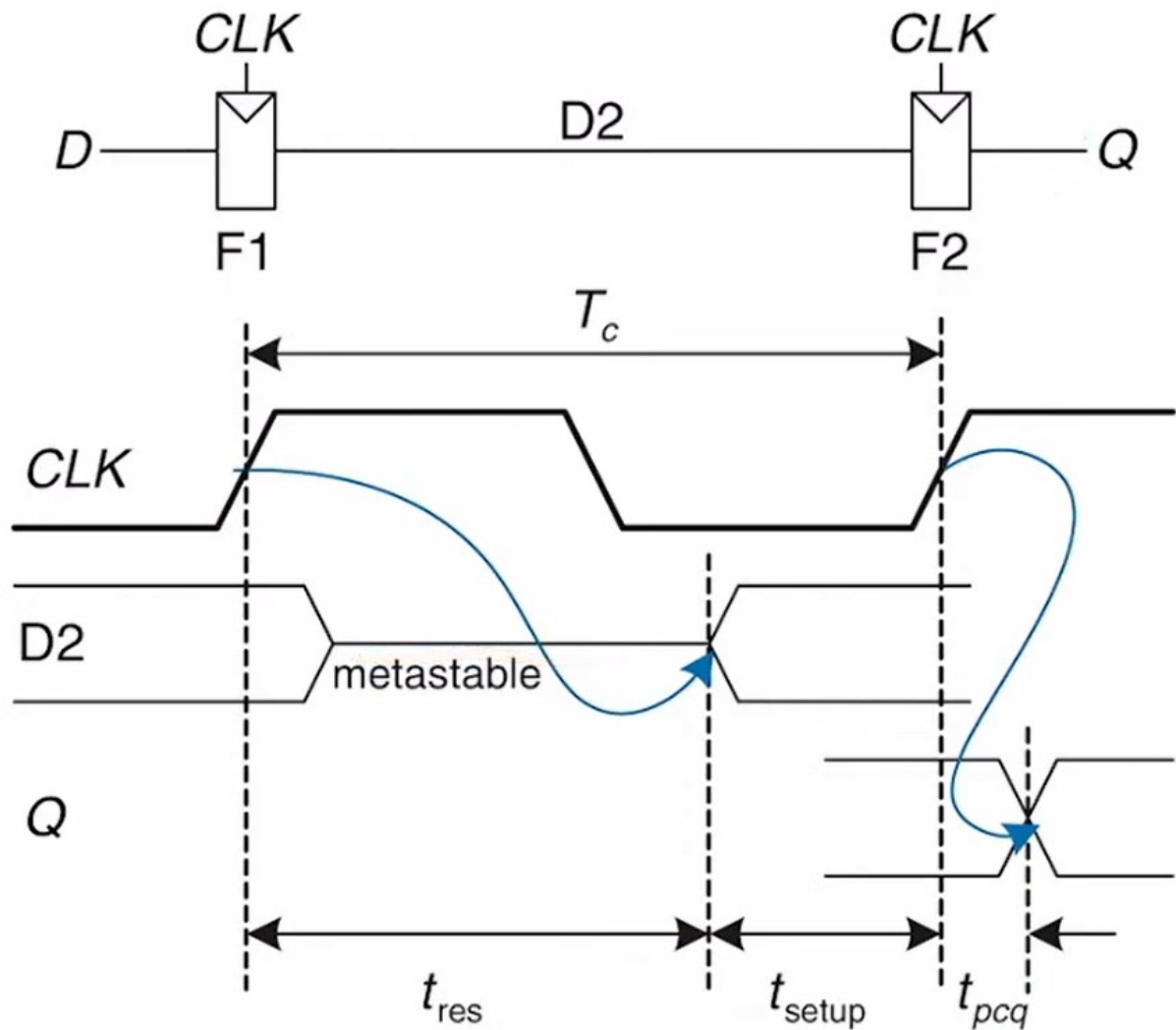
Synchronizers

To deal with above problem, we use synchronizers which are highly likely to give either 0 or 1, no meta-state



i.e. 2 DFF's, but it causes us delay like 2 clock cycle or more

Synchronizers



- $t_{res} + t_{setup} < T_c$

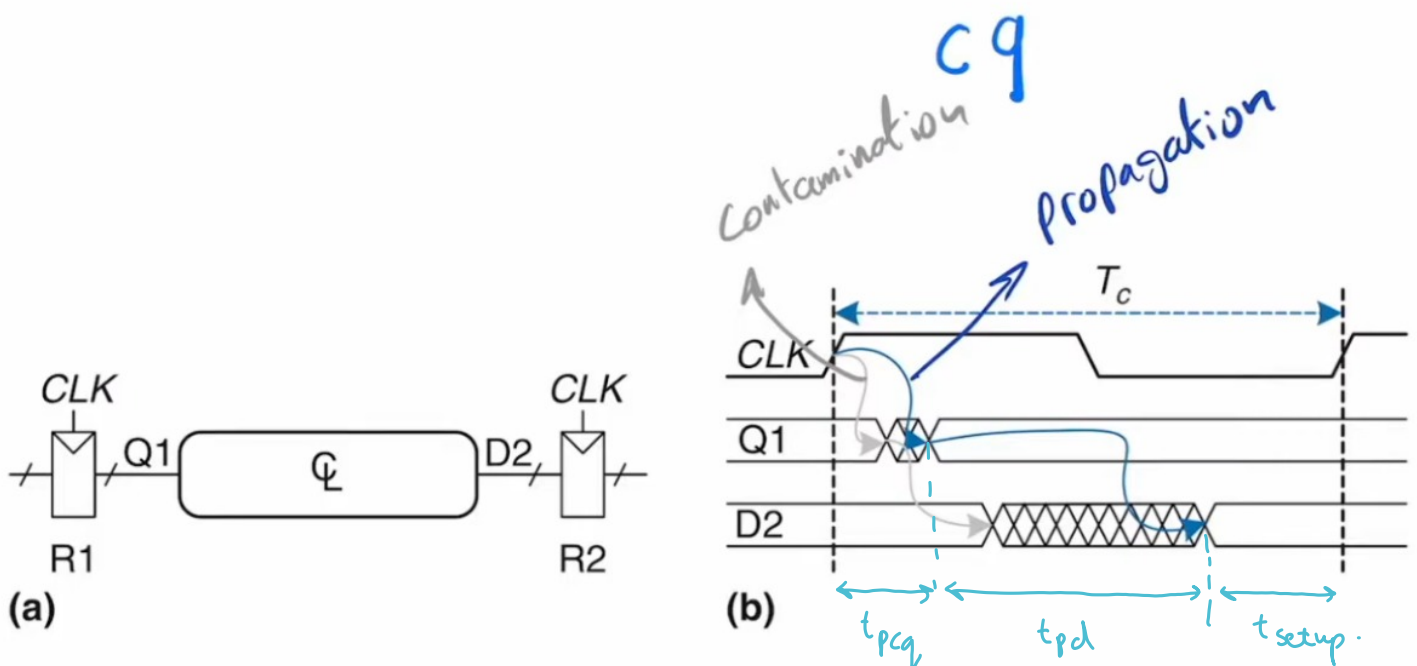
- we can add more f.f.'s but it will cost us delay, so tradeoff

Sequential Circuit Timing Analysis

① Analysis Paths

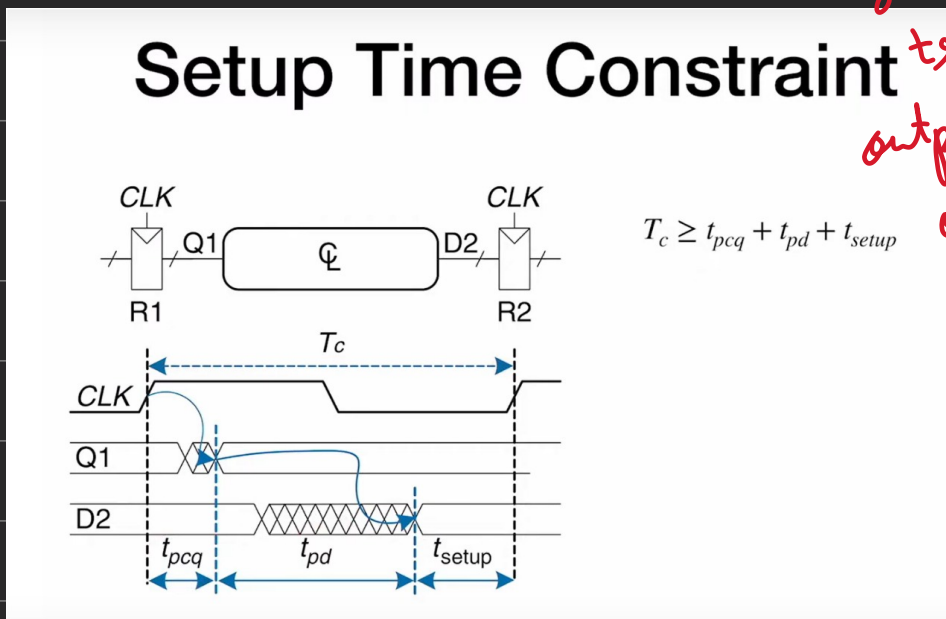
- Register to register (i.e. F.F. to F.F.)
- Primary input to register
- Register to primary output
- Input to output (i.e. no. F.F.)

Register to Register



propagation delay

Setup Time Constraint



after clk edge triggers, the output Q takes some time to change

↳ Setup Time constraint

$$T_c \geq t_{pcq} + t_{pd} + t_{setup}$$

by last value

$$f_c = \frac{1}{T_c}$$

fixed.

we usually have pd of that comb. circuit

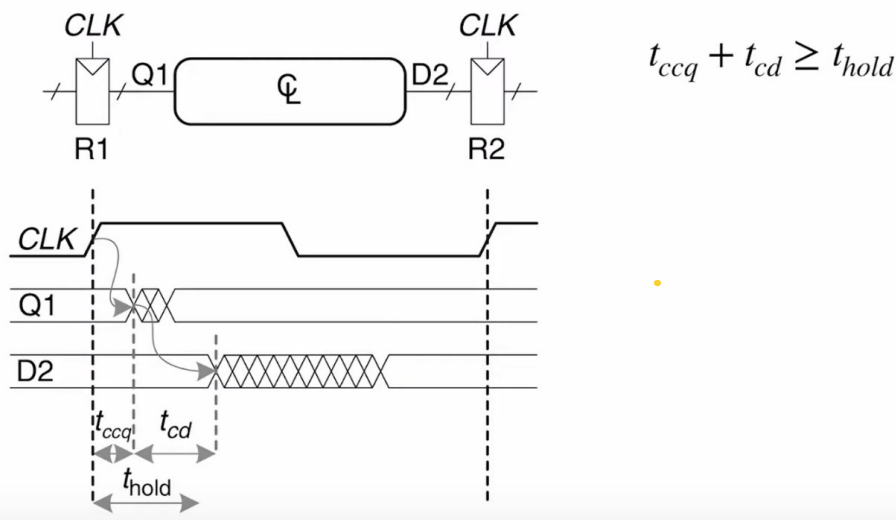
$$\Rightarrow t_{pd} \leq T_c - (t_{pcq} + t_{setup})$$

sequential overhead

"slack is the time left before violating time constraint"

with
+ve is preferred, -ve is bad

Hold Time Constraint



↳ Hold time constraint

$$t_{ccq} + t_{cd} \geq t_{hold}$$

only thing in our hand

$t_{cd} \rightarrow$ min time out of that combinational logic

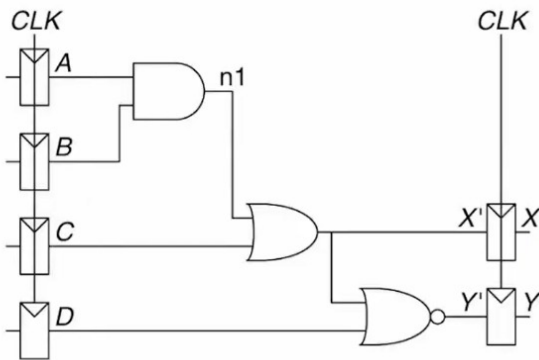
hold time doesn't affect our max. freq.ency.

$$\Rightarrow \begin{aligned} t_{cd} &\geq t_{hold} - t_{cq} \\ t_{pd} &\leq T_c - (t_{pcq} + t_{setup}) \end{aligned}$$

min & max delay of our C.L. should be within this range.

for Good F.F. $t_{hold} \leq t_{ccq}$, so hold time violation never occurs even if t_{cd} is 0

Timing Analysis Example



Flip-flops

$$t_{ccq} = 30 \text{ ps} \quad t_{setup} = 50 \text{ ps}$$

$$t_{pcq} = 80 \text{ ps} \quad t_{hold} = 60 \text{ ps}$$

Each logic gates:

$$t_{cd} = 25 \text{ ps} \quad t_{pd} = 40 \text{ ps}$$

contaminated delay

check for violation

$$T_c \geq t_{pcq} + t_{pd} + t_{setup} - \textcircled{1}$$

$$T_{hold} \leq t_{ccq} + t_{cd} - \textcircled{2}$$

~~B/A - X' → 60ps ≤ 30 + 50 ✓~~
no hold time violation.

$$T_c \geq 80 + 80 + 50$$

~~B/A - Y' → 66ps ≤ 30 + 75 ✓~~

$$C - X' \rightarrow \begin{aligned} T_c &\geq 80 + 120 + 50 \\ 60 \text{ ps} &\leq 30 + 25 \quad \times \\ T_c &\geq 80 + 10 + 50 \end{aligned}$$

$$C - Y' \rightarrow \begin{aligned} 60 \text{ ps} &\leq 30 + 50 \quad \checkmark \\ T_c &\geq 80 + 80 + 50 \end{aligned}$$

$$D - Y' \rightarrow \begin{aligned} 60 \text{ ps} &\leq 30 + 25 \quad \times \\ T_c &\geq 80 + 10 + 50 \end{aligned}$$

will not be done this way

$$T_c \geq t_{prq} + t_{pd} + t_{setup}$$

non possible

3F.F.

$$T_c \geq 80 + 3 \times 40 + 50$$

$$f_c \leq \frac{1}{250 \text{ ps}} \leq 4 \text{ GHz}$$

$$T_{hold} \leq t_{ccd} + t_{cd} \rightarrow \text{min}$$

$$60 \text{ ps} \leq 30 \text{ ps} + 25$$

violated

* Add buffer (min time path) to remove hold time violation

Vivada Timing Report

1.

① Worst Negative slack (WNS)

after checking whole circuit
gives worst slack (min time
before you violate timing
constraint)

+ve preferred

② Total Negative slack (TNS)

adds all neg slack & gives
us.

0 preferred.

③ Worst Hold slack (WHS)

↳ during hold check

④ Total Hold Slack (THS)

clock skew

possible cause

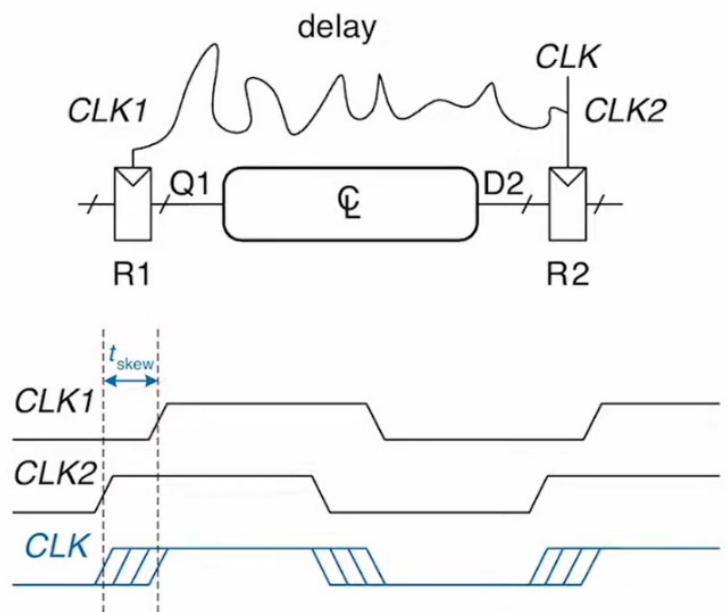
- Different clock path lengths
- Noise on clock path
- Clock gating

Clock Skew

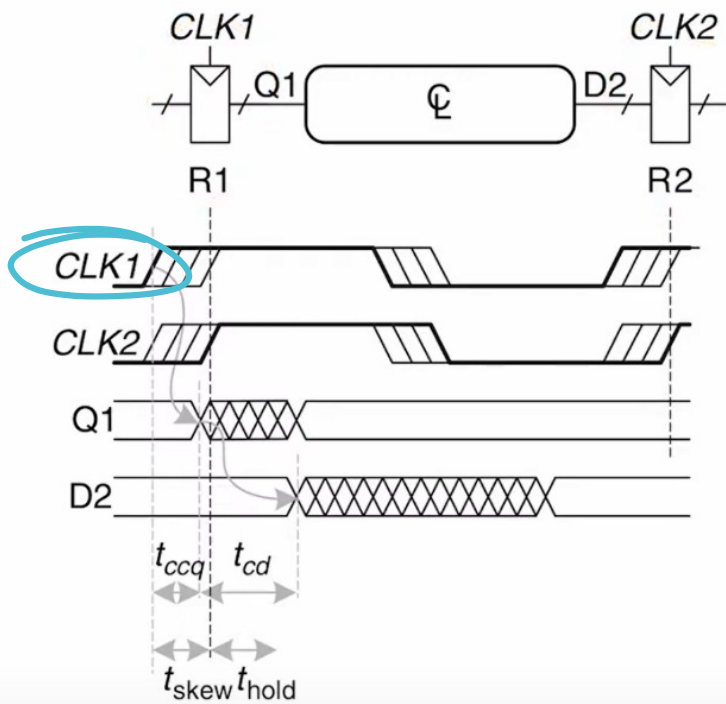
Different clock path lengths

Noise on the clock path

Clock gating



• 'if contamination delay' which is earliest



$$t_{ccq} + t_{cd} \geq t_{hold} + t_{skew}$$

$$t_{cd} \geq t_{hold} + t_{skew} - t_{ccq}$$

- contamination delay $\rightarrow$ hold time
 $\hookrightarrow$ first clk

- $\hookrightarrow$ we need to reduce clock skew
 & we do it by using resources
 from the manufacturer of the
 chip
 $\hookrightarrow$ do not generate your
 own clock, $\rightarrow$ CAD tool
 may not understand it as
 clock signal & might not
 used clock trees.

T_{pd}



#

1) $T_{clk \rightarrow q} + T_{comb} + T_{setup} \leq T_{clk} + T_{skew}$

clock skew (+ve skew helps setup) ←

clk to Q delay of launching f.f.

Setup time requirement of capturing f.f.

T_{cd}



2) $T_{clk \rightarrow q} + T_{comb} > T_{hold} + T_{skew}$

→ Setup constrain ensures data arrives & is stable before clock edge.

→ Hold constrain ensures data remain stable after the clock edge.

→ +ve clock skew helps setup timing but hurts hold timing.

→ Setup violation are fixed by reducing combinational delay or increasing clock frequency.

→ Hold violation are typically fixed by adding buffer delays in data path.

